



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



NASA Core Flight System Health and Safety Application Vulnerability Exposure

Threat Report

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TLP:CLEAR | Lost Boys Cyber | NASA Core Flight System Health and Safety Application Vulnerability Exposure - Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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NASA Core Flight System Health and Safety Application Vulnerability Exposure

1. Executive Summary

CISA ICS advisory ICSA-26-211-06 identifies a NASA Core Flight System Health & Safety Application vulnerability affecting versions $\leq v7.0.1$ and states successful exploitation could cause a denial-of-service condition. This is a space-system software exposure rather than a generic enterprise vulnerability, so aerospace programs should treat inventory confirmation and patch planning as sector-specific risk work.

This automated daily report favors practical defender action over speculative attribution. It uses public sources available during the 2026-09-12 UTC run and clearly scopes older sector-relevant sources where current reporting was limited.

2. Intelligence Requirements

Question	Current Answer	Confidence
What changed or remains operationally relevant today?	The defender priority is to confirm whether cFS HS components are used in lab, simulation, ground, or operational environments and ensure access paths to relevant command/telemetry interfaces are tightly controlled.	Medium
Which environments should prioritize review?	Space systems, satellite software labs, aerospace integrators, ground-system operators, and research environments using NASA cFS components.	Medium
Are hard IOCs available from public sources?	No unique IOCs were safely extracted in this run; use behavior-based hunts and source follow-up.	Medium

3. Source Review & Confidence

Source	Contribution	Confidence
CISA ICSA-26-211-06 NASA cFS HS Application	Primary advisory for CVE-2026-18064 and affected NASA cFS HS versions.	High
NASA Core Flight System project	Project context for NASA Core Flight System.	Medium
CISA ICS advisories index	Advisory stream for ICS/OT and embedded system vulnerabilities.	Medium

4. Threat Activity Overview

The defender priority is to confirm whether cFS HS components are used in lab, simulation, ground, or operational environments and ensure access paths to relevant command/telemetry interfaces are tightly controlled.

5. Affected Sectors and Exposure

Space systems, satellite software labs, aerospace integrators, ground-system operators, and research environments using NASA cFS components.

Exposure Pattern	Operational Concern
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Identity compromise	Enables remote access, SaaS abuse, mailbox/data theft, or supplier impersonation.
Exposed internet-facing platforms	Increases scan/exploit likelihood and shortens defender response window.
Third-party or supply-chain dependency	One provider incident can create multi-organization impact.
Weak egress/data monitoring	Reduces ability to detect staging, exfiltration, or C2.

6. Aviation and Aerospace Sector Impact

Sector Element	Analyst Assessment
Affected segment	satellite/space software and aerospace integration environments
Operational relevance	Segment: satellite/space software and aerospace integration environments. Operational relevance is availability and health/safety monitoring integrity. Exposure paths include lab networks, ground systems, simulation ranges, and contractor-maintained software baselines.
Likely exposure paths	Identity compromise, supplier remote access, shared platforms, exposed management interfaces, and weak segmentation between corporate IT and operations-support environments.
Third-party / supply-chain considerations	Require supplier patch attestations, incident notification SLAs, remote-access reviews, and tabletop exercises for service disruption.
Defender actions for aviation operators	Inventory dependencies, validate backups and fallbacks, strengthen MFA/conditional access, monitor privileged accounts, and rehearse degraded-mode operations.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed or Plausible Technique	Evidence
Initial Access	T1566 Phishing or T1190 Exploit Public-Facing Application	Source-dependent: document lures, external services, or exposed platforms are the common entry themes.
Execution	T1059 Command and Scripting Interpreter	Hunt logic focuses on script/command execution after suspicious access.
Collection / Exfiltration	T1560 Archive Collected Data; T1567 Exfiltration Over Web Service	Relevant to extortion, espionage, and cloud-enabled operations.
Impact	T1486 Data Encrypted for Impact / service disruption	Applies where ransomware or platform disruption is the reported risk.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
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No durable atomic IOCs extracted	Source limitation	Do not block on invented IOCs; hunt for behaviors and affected technologies.
Named technologies/sectors in this report	Asset/sector pivot	Use CMDB, EDR tags, external attack-surface management, and supplier inventories to scope exposure.

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
New inbound or management-plane access to priority exposed system	Firewall, VPN, WAF, reverse proxy, cloud flow logs	Identify opportunistic exploitation or reconnaissance against the named technology/theme
Suspicious child process, script execution, or anomalous service-account action near affected asset	EDR, server process logs, audit logs	Find post-exploitation activity after initial access
Archive creation, bulk reads, token changes, or unusual outbound transfer	EDR, DLP, proxy, cloud audit logs	Detect staging and exfiltration attempts tied to the reported activity

9.2. Sigma / Detection Content

<pre> title: LBC Daily Priority Suspicious Server Post Exploitation - nasa-core-flight-system-health-safety-application-exposure id: lbc-nasa-core-flight-system-health-safety-ap status: experimental description: Flags suspicious command execution and staging behavior on servers or edge systems associated with this daily priority. Tune asset filters before production use. logsource: category: process_creation product: windows detection: selection_img: Image endswith: - '\\cmd.exe' - '\\powershell.exe' - '\\pwsh.exe' - '\\wscript.exe' - '\\cscript.exe' selection_cmd: CommandLine contains: - 'curl ' - 'wget ' - 'Invoke-WebRequest' - 'certutil -urlcache' - 'tar ' - '7z ' - 'rclone' condition: selection_img and selection_cmd falsepositives: - Administrative maintenance scripts on managed servers level: high </pre>
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9.3. Hunt Query

<pre> DeviceProcessEvents where Timestamp > ago(14d) where FileName in~ ("cmd.exe", "powershell.exe", "pwsh.exe", "wscript.exe", "cscript.exe") where ProcessCommandLine has_any ("curl ", "wget ", "Invoke-WebRequest", "certutil -urlcache", </pre>
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```
"rclone", "7z", "tar ")
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Count=count(),
Commands=make_set(ProcessCommandLine, 5) by DeviceName, InitiatingProcessAccountName, FileName
| order by LastSeen desc
```

10. Threat Hunting

10.1. Hunt Hypotheses

Hypothesis	Data Required	Expected Finding
Exposed or high-value systems tied to this report show exploitation or reconnaissance within the last 14 days.	Firewall, WAF, reverse proxy, VPN, cloud flow logs	Spikes in requests, failed authentication, rare user agents, or unexpected source geographies
A compromised server or workstation staged data or tooling after initial access.	EDR process/file/network events	Archive tools, scripted downloads, web shells, unusual outbound transfer, or rare parent/child process chains
<pre>DeviceNetworkEvents where Timestamp > ago(14d) where RemoteUrl has_any ("pastebin", "githubusercontent", "workers.dev", "pages.dev", "ngrok", "trycloudflare") or RemotePort in (4444, 8080, 8443, 9001) summarize Connections=count(), FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Urls=make_set(RemoteUrl, 10) by DeviceName, InitiatingProcessFileName, RemoteIP, RemotePort order by LastSeen desc</pre>		

11. Risk Assessment

Dimension	Assessment
Likelihood	High for organizations matching the exposure profile; lower for segmented/non-exposed environments.
Impact	Potential credential theft, data exfiltration, service disruption, or follow-on intrusion depending on theme.
Confidence	Medium based on source specificity and recency.

12. Recommended Actions

Priority	Owner	Action
Critical	SOC lead	Run the hunt queries against in-scope assets and review hits with source context.
High	Platform / identity teams	Validate MFA, patch state, service-account governance, and logging for exposed systems.
High	Incident response	Prepare containment playbooks for the named sector/technology scenario.
Medium	Vendor management	Request supplier status for relevant platforms and critical service dependencies.

13. References

- [1] CISA ICSA-26-211-06 NASA cFS HS Application: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-211-06> — Primary advisory for CVE-2026-18064 and affected NASA cFS HS versions.

- [2] NASA Core Flight System project: <https://cfs.gsfc.nasa.gov/> — Project context for NASA Core Flight System.
- [3] CISA ICS advisories index: <https://www.cisa.gov/news-events/ics-advisories> — Advisory stream for ICS/OT and embedded system vulnerabilities.